Experiment 1

Name: Rishi Jain
UID: 23BAI70569
Branch: BECSE-AIML
Section/Group: 23AIT_KRG-1

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1. **AIM:** To design and implement the foundational frontend architecture of the EcoTrack application using modern React practices, Vite tooling, and ES6+ JavaScript features.
2. **Objective:**

a. To set up a React project using Vite with proper project structure
b. To understand component-based architecture in React
c. To apply ES6 array methods (map, filter, reduce) for data-driven UI rendering
d. To separate concerns using components, pages, and data modules

3. **Implementation/Code:**

a.Dashboard.jsx

```
import logs from "../data/logs";

const Dashboard = () => {
const sum = logs.reduce((acc, ele) => acc + ele.carbon, 0);

return (
<div
style={{
display: "flex",
justifyContent: "center",
marginTop: "40px"
}}
>
{/* TOTAL CARBON CARD ONLY */}
<div
```

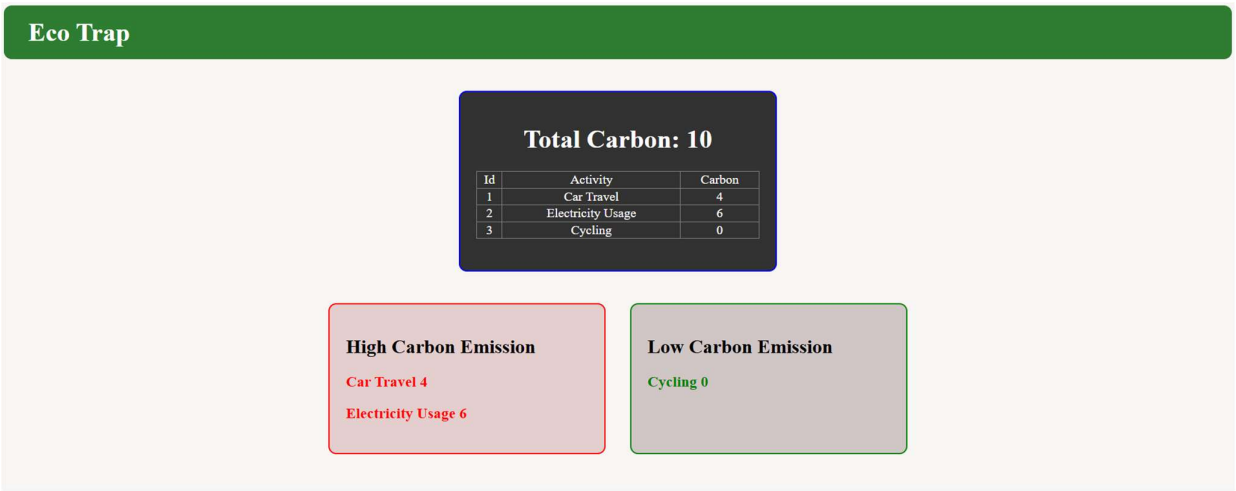
```
style={{
backgroundColor: "#323232ff",
width: "350px",
padding: 20,
borderRadius: 10,
border: "2px solid blue",
color: "#fff",
textAlign: "center"
}}
>
<h1>Total Carbon: {sum}</h1>

<table
style={{
margin: "20px auto",
borderCollapse: "collapse",
width: "100%",
color: "#fff",
textAlign: "center"
}}
border="1"
>
<tr>
<td>Id</td>
<td>Activity</td>
<td>Carbon</td>
</tr>

{logs.map((element) => (
<tr key={element.id}>
<td>{element.id}</td>
<td>{element.activity}</td>
<td>{element.carbon}</td>
</tr>
))}
</table>
</div>
</div>
);
};

export default Dashboard;
```

b. Logs.jsx:

```
import logs from "../data/logs";

const Logs = () => {
return (
<div
style={{
display: "flex",
justifyContent: "center",
gap: "30px",
marginTop: "40px",
flexWrap: "wrap" // responsive
}}
>
{/* HIGH CARBON */}
<div
style={{
backgroundColor: "rgb(229, 206, 206)",
width: "300px",
padding: 20,
borderRadius: 10,
border: "2px solid red"
}}
>
<h2>High Carbon Emission</h2>
{logs
.filter((ele) => ele.carbon >= 4)
.map((ele) => (
<h3 key={ele.id} style={{ color: "red" }}>
{ele.activity} {ele.carbon}
</h3>
))}
</div>

{/* LOW CARBON */}
<div
style={{
backgroundColor: "rgb(209, 198, 198)",
width: "300px",
padding: 20,
borderRadius: 10,
border: "2px solid green"
}}
>
<h2>Low Carbon Emission</h2>
{logs
.filter((ele) => ele.carbon < 4)
.map((ele) => (
<h3 key={ele.id} style={{ color: "green" }}>
```

```
{ele.activity} {ele.carbon}
</h3>
))}
</div>
</div>
);
};

export default Logs;
```

4.Output:



5. Learning Outcome:

a. I learnt how to make react applications.
b. How the react dom actually works.
c. I learnt how to use the filter, map and the reduce function.